Machine Learning in Time Series

Program and Lecture 1

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Agenda

1. Seminar Overview & Mechanics
2. Lecture 1: Historical Evolution, the Prediction Problem, and Shrinkage
3. The Epistemological Shift
4. The Prediction Problem & Evaluation
5. The Curse of Dimensionality & Overfitting
6. Regularization: Ridge and Lasso
7. Readings & Lecture 1 Mini-Lab
8. Mini-Lab: Empirical Results & Commentary
9. Concluding Remarks

Seminar Overview & Mechanics

Seminar Objective

The Core Objective

This seminar studies the interface between traditional time-series econometrics and modern machine learning (ML) for (i) forecasting and (ii) policy-relevant inference.

Central Themes:

- Treating ML methods as explicit optimization/statistical objects, not black boxes.
- Proper evaluation under strict time-series constraints: serial dependence, nonstationarity, structural breaks, and data revisions.
- Shifting safely from structural identification to algorithmic risk minimization.

Course Mechanics & Evaluation

Format

- 6 Lectures, 3 hours each.
- **Structure per Lecture:** Theory & derivations ($\sim$75 min), Paper discussion ($\sim$75 min), and a Coding lab component ($\sim$30 min).
- **Tools:** Python or R.

Evaluation Metrics

- **25% - Memos & Participation:** 1-page reading memo due before each Lecture.
- **25% - Presentation:** Present one syllabus paper + a 10-minute code walkthrough.
- **50% - Final Project:** 10-15 page replication paper or research proposal. Includes a complete empirical pipeline and honest OOS evaluation.

The Seminar Project

A Fixed Evaluation Protocol

To compare methods fairly without getting lost in "cool models", the entire 6-week seminar runs on a single canonical forecasting task. Every new model answers the same question under the exact same rules.

The Task: Multi-horizon forecasting of YoY Core CPI Inflation.

- **Baselines:** Mean/AR, small VAR, (optionally BVAR).
- **Data-Rich Extension:** Later lectures will inject FRED-style predictor panels.
- **Evaluation Rule:** Strict rolling/expanding OOS, leakage-safe tuning.
- **Horizons:** $h = \{1, 3, 6, 12\}$.
- **Metrics / Tests:**
  - Forecast loss: RMSE / MAE by horizon.
  - Comparison: Diebold-Mariano test when appropriate.
  - Classification tasks: AUC / PR + calibration.

The 6-Lectures Roadmap

Lecture 1: Historical Evolution, the Prediction Problem, and Shrinkage.

Lecture 2: Kernel Methods and Support Vector Machines (SVM) for Time Series.

Lecture 3: High-Dimensional Linear Dynamics (BVAR, Sparse VAR, & Structured Regularization).

Lecture 4: Deep Learning for Sequences (RNN/LSTM and Attention-Based Forecasting).

Lecture 5: Trees, Boosting, and Ensembles for Nonlinear Forecasting.

Lecture 6: The Frontier: Attention, Text, and Causal ML.

Lecture 1: Historical Evolution, the Prediction Problem, and Shrinkage

The Epistemological Shift

Classical Econometrics vs. Machine Learning

Classical Econometrics

Goal: Recover the true DGP
Target: $\mathbb{E}[Y|do(X)]$ (Causal)
Setting: $T \gg p$ (Low-dimensional)

Focuses on unbiasedness, parameter identification, and structural policy invariance (Lucas Critique).

The Algorithmic Turn

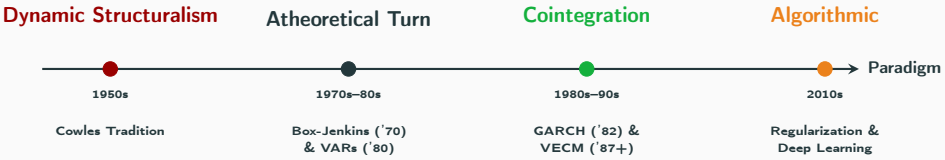
Goal: Minimize OOS Loss
Target: $\hat{f}(X) \approx \mathbb{E}[Y|X]$ (Predictive)
Setting: $p \approx T$ or $p > T$

Focuses on generalization, variance reduction via shrinkage, and handling massive, unstructured data.

The Fourth Revolution (Villena, 2026)

ML is not a rejection of econometrics; it is an evolution forced by the data explosion, trading parameter interpretability for forecast stability.

Historical Context: The Four Revolutions

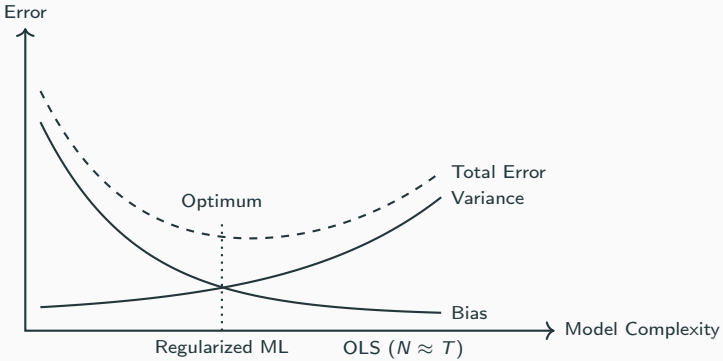


Note: This is a highly stylized heuristic timeline. Each phase emerged to solve the empirical failures of the previous paradigm (e.g., Sims' critique leading to VARs).

The Prediction Problem & Evaluation

The Bias-Variance Tradeoff

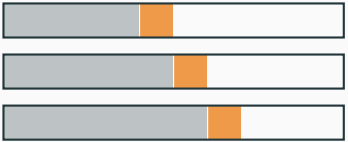
In modern macro, OLS yields Bias ≈ 0, but Variance explodes as predictors $p \to T$. ML algorithms actively introduce bias to suppress this variance.



Evaluating Time Series: Avoiding Data Leakage

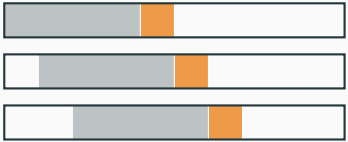
Standard K-fold cross-validation fails in time series due to temporal dependence. We must use strict sequential splits to mirror real-world forecasting.

Expanding Window (Recursive)



Mirrors a central bank utilizing all historical data. Best for stationary series.

Rolling Window



Fixed training size. Best when older data represents a discarded structural regime.

The Curse of Dimensionality & Overfitting

The Limits of Classical OLS

What breaks OLS?

Ordinary Least Squares fits a hyperplane by strictly minimizing in-sample MSE.

Vulnerabilities in Modern Macro:

- Highly sensitive to structural breaks and outliers.
- Cannot handle severe multicollinearity (singular $X^\top X$).
- **Fails completely** when predictors outnumber observations ($p \gg T$).

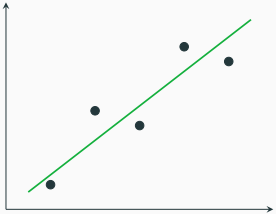
The Curse of Dimensionality

As we add more variables (FRED-MD, text data), the mathematical volume of the data space grows exponentially.

Data points become extremely sparse. OLS reacts by assigning massive, unstable coefficients to "chase" the noise, destroying Out-of-Sample (OOS) predictive power.

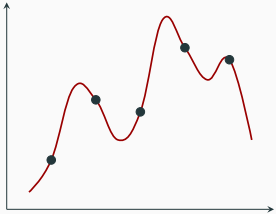
Visualizing the Problem: Overfitting

Good Fit (Learned the DGP)



The model captures the general structural trend. It performs well on unseen rolling windows.

Overfitting (Memorized the Noise)



Like memorizing a practice test. $R^2 \approx 1$ in-sample, but MSE explodes out-of-sample.

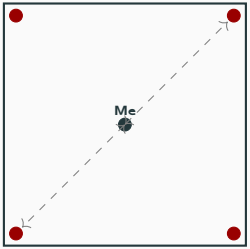
Why Classic Rules Break: The Failure of Distance

The "Magic Playground"

Imagine playing tag in a room with **1,000 directions** to run. Because there is so much empty space, everyone runs into their own corner.

Result: Everyone is exactly 1 mile away from you. If everyone is the same distance, the concept of "closest" disappears!

High-D Playground (Everyone is in a corner)



When algorithms measure distance between 10,000 variables, the measurements become mathematically useless.

Regularization: Ridge and Lasso

The Solution: Regularization (Shrinkage)

To survive high-dimensional environments ($p \gg T$), we must enforce **mathematical discipline** to control bias.

Instead of just minimizing empirical error, we minimize the error **PLUS** a penalty for complexity:

The Penalized Loss Function

$$\hat{\beta} = \arg\min_{\beta} \{\mathcal{L}(Y, X\beta) + \lambda \cdot \mathcal{P}(\beta)\}$$

- $\lambda \geq 0$ is the tuning parameter controlling the severity of the penalty.
- By forcing the model to generate smaller parameters, we trade a small amount of **bias** for a massive reduction in **variance**.

Ridge Regression (L_2 Regularization)

Ridge Regression adds a squared Euclidean norm penalty to the objective:

$$\hat{\beta}_{Ridge} = \arg\min_{\beta} \left\{ \frac{1}{2T} \|Y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}$$

Key Properties:

- **Smooth Shrinkage:** Pulls coefficients toward zero smoothly, but rarely reaches exactly zero.
- **Analytic Solution:** Easily solved via linear algebra: $\hat{\beta}_{Ridge} = (X^\top X + \lambda I)^{-1} X^\top Y$.
- **Econometric Link:** Mathematically equivalent to a Bayesian VAR (BVAR) with a Gaussian prior centered at zero (e.g., Minnesota Prior).

Lasso: Penalized vs. Constrained Formulation

The Lasso (L_1) estimator can be mathematically expressed in two equivalent forms:

1. The Penalized Form:

$$\min_{\beta} \left\{ \frac{1}{2T} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

Here, $\lambda \geq 0$ acts as the penalty weight controlling the shrinkage intensity.

2. The Constrained Form:

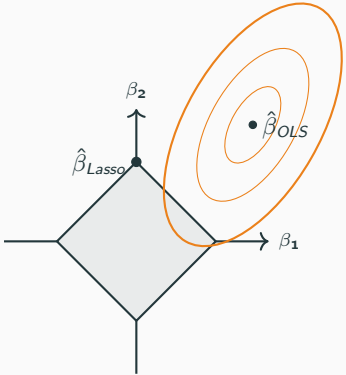
$$\min_{\beta} \left\{ \frac{1}{2T} \|y - X\beta\|_2^2 \right\} \quad \text{s.t.} \quad \|\beta\|_1 \leq t$$

Here, $t \geq 0$ acts as an absolute "budget" for the coefficients. The KKT conditions establish that for every budget t, there exists a corresponding penalty λ.

The Geometry of Shrinkage

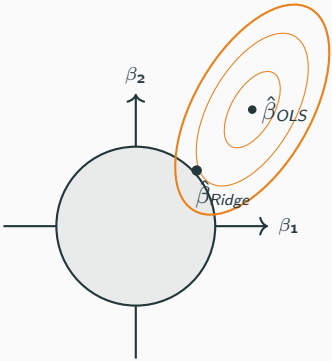
Why does Lasso select variables ($\beta_j = 0$) while Ridge does not?

Lasso (L_1 Diamond)



The sharp corners mean the contour hits exactly on an axis ($\beta_1 = 0$).

Ridge (L_2 Circle)



The circular boundary intersects off the axis, shrinking both without eliminating either.

The Mathematical Logic of the Penalties

Let's look at a numerical example to see how the penalties behave differently.

Suppose our current coefficients are $\beta = [10, 5]$. What happens if the optimizer reduces the first parameter by exactly 1 unit (to 9)?

Scenario	**Ridge (L_2) Penalty ($\sum \beta_i^2$)**	**Lasso (L_1) Penalty ($\sum \lvert\beta_i\rvert$)**
Initial: $[10, 5]$	$10^2 + 5^2 = 125$	$10 + 5 = 15$
New: $[9, 5]$	$9^2 + 5^2 = 106$	$9 + 5 = 14$
Reduction Size	**-19**	**-1**

Insight

Ridge prioritizes shrinking **large** parameters because squaring large numbers increases the penalty exponentially. Lasso treats all unit reductions equally, allowing it to push a parameter to exactly zero.

Summary Comparison

Feature	Ridge (L_2)	Lasso (L_1)
Primary Goal	Handle multicollinearity & stabilize	Feature selection & simplicity
Resulting Model	All features included (small weights)	Sparse model (irrelevant features = 0)
Computational Solution	Closed-form (Matrix algebra)	Numerical optimization (Iterative)
Bias-Variance Tradeoff	Lower variance, slightly higher bias	Lower variance, potentially higher bias

Both models are essential tools for modern econometricians navigating data-rich ($p \gg T$) environments!

Readings & Lecture 1 Mini-Lab

Required & Recommended Readings

- **Villena (2026):** *From Lagged Effects to Algorithmic Prediction*. Sets the historical framing for the "Four Revolutions" and the shift toward algorithmic prediction.
- **Masini, Medeiros, & Mendes (2023):** *Machine learning advances for time series forecasting*. Crucial for understanding evaluation under temporal dependence.
- **De Mol, Giannone, & Reichlin (2008) [Recommended]:** *Forecasting using a large number of predictors...* Demonstrates that when data is highly collinear, Ridge regression and factor models (e.g. PCA - Select the top factors that explain a significant portion of variance.) are deeply connected and solve the curse of dimensionality effectively.

Lecture 1 Mini-Lab: Leakage-Safe Shrinkage

The Project Task

Build a leakage-safe Ridge/Lasso forecast for US YoY CPI (our canonical target) and compare it to a classical OLS benchmark using provided starter code - YoY Year over Year, 12-month change.

Two R Scripts are provided for this lab:

1. **Script 1 (Baseline):** A static train-test split to introduce the `glmnet` syntax and AR(p) feature engineering.
2. **Script 2 (The Protocol):** A rigorous h-step ahead direct forecasting loop using rolling windows across multiple horizons.

Understanding Script 1: The Baseline AR(12) Pipeline

Core Mechanics of the Script:

- **Feature Construction:** We use `embed(inflation, p + 1)` to automatically construct the X matrix of $t-1 \dots t-12$ lags without manual looping.
- **Temporal Split:** We strictly split the data at the 80% mark (`1:split_index`). *No random sampling is used* to prevent future information leaking into the training set.
- **Tuning λ:** The function `cv.glmnet(x_train, y_train)` performs k-fold cross-validation to find `lambda.min`.
- **Important Note:** We use standard k-fold here *only* to introduce the syntax. Because random k-fold breaks temporal order, all substantive results must use rolling/blocked CV or a time-ordered validation scheme!

Understanding Script 2: Rolling Windows & Direct Forecast

The second script implements our Seminar Project evaluation protocol:

1. **Direct h-step Ahead Forecasting:** Instead of forecasting $t+1$ and iterating iteratively to $t+h$, we model y_{t+h} directly as a function of X_t. The script shifts the feature index back by h:
   - `train_feature_idx <- train_target_idx - h`
2. **The Rolling Loop:**
   - **Outer Loops:** Iterate over forecast horizons ($h = 1, 3, 6, 12$) and window sizes ($T = 60, 120$ months).
   - **Inner Loop (`step in 1:OOS_steps`):** Moves the training window forward exactly one month at a time, retraining the OLS, Ridge, and Lasso models at every single step before predicting exactly one out-of-sample data point.

Mini-Lab: Empirical Results & Commentary

Lab Results: The Baseline AR(12) Split

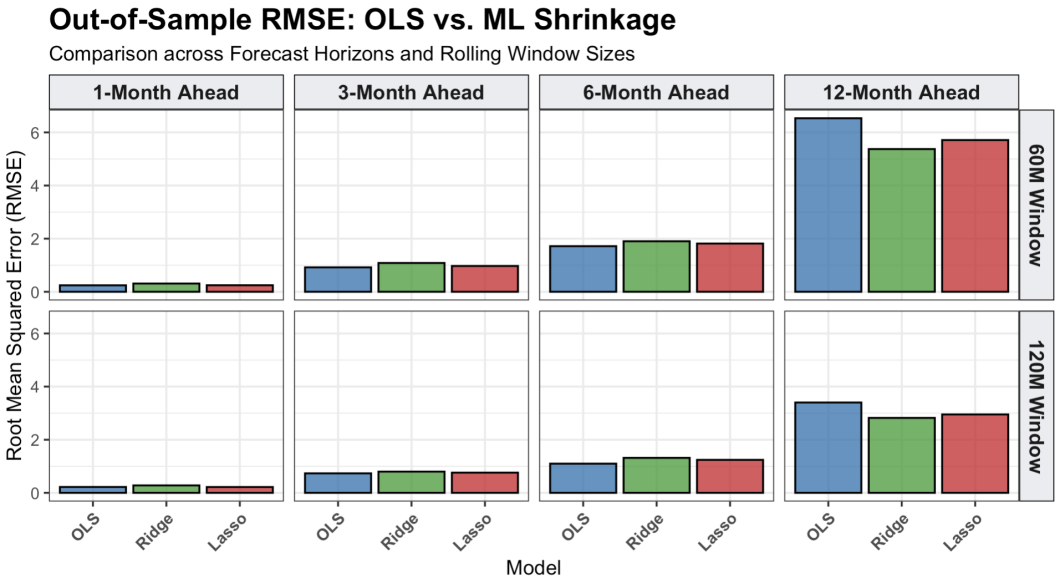
Model	RMSE
Standard AR(12) (OLS)	0.1865
Lasso AR(12) (L1)	0.1959
Ridge AR(12) (L2)	0.3297

Note: Baseline RMSE uses k-fold tuning for illustration; do not interpret it as leakage-robust.

Why did OLS win?

In a low-dimensional setting where the model is well-specified and variance is already moderate, shrinkage can add bias *without* providing enough variance reduction to improve OOS RMSE.

Lab Results: Rolling Windows & Long Horizons



Lab Results: Rolling Windows & Long Horizons

When Machine Learning saves the forecast:

- **Long Horizons ($h = 12$):** Signal-to-noise ratio drops. OLS overfits to noise, causing RMSE to explode.
- **Small Windows ($T = 60$):** With only 5 observations per parameter, OLS variance becomes unstable.
- **The ML Edge:** Ridge and Lasso act as "insurance." By penalizing large coefficients, they stabilize the model during volatile periods or when degrees of freedom are scarce.

Takeaway: Shrinkage is a requirement for stability, not just an option.

Is Inflation "Linear" and "Stationary"? Implications for Forecasting and Shrinkage

Key point. Treating YoY CPI inflation as strictly linear and stationary is often a useful *local approximation*, but it is not a safe global assumption for forecasting or policy inference.

1) Stationarity: what is plausibly true.

- Monthly CPI levels are nonstationary; YoY inflation is *more* stable and can look stationary in long samples.
- Still, inflation exhibits **persistent dynamics**, **time-varying volatility** and **structural breaks** (policy regimes, supply shocks, measurement/revisions).

2) Linearity: what is plausibly true.

- An AR(12) is a linear conditional-mean model; it can fit well when dynamics are stable.
- Nonlinearities can arise from **regime dependence** (high vs low inflation), **asymmetric responses** to shocks, and **changing persistence**.

Is Inflation "Linear" and "Stationary"? Implications for Forecasting and Shrinkage

3) Why OLS may beat Ridge/Lasso in your AR(12).

- With only $p = 12$ regressors and decent T, OLS variance is already low; shrinkage adds bias and may not reduce RMSE.
- If λ is tuned with random k-fold CV, time-order is violated and results can be distorted; use rolling/blocked tuning.

4) What to do in practice (seminar rule).

- Diagnose: unit-root tests + break tests; check stability with rolling estimates; examine residual autocorrelation and heteroskedasticity.
- Evaluate honestly: rolling/expanding OOS, multi-horizon loss (RMSE/MAE), and (when relevant) forecast comparison tests.
- Expect ML gains mainly in **data-rich**, **unstable**, or **nonlinear** environments (many predictors, regime shifts, interactions).

Concluding Remarks

Project: 10-Minute Check-In

Our Standard Closing Questions

1. **What changed in OOS RMSE by horizon?**
2. **Which failure mode did we fix (or create)?**

Today's Answers:

- **RMSE Shift:** OLS dominated $h = 1$ and large windows ($T = 120$). ML Shrinkage drastically improved $h = 12$ on small windows ($T = 60$).
- **Failure Mode Fixed:** We fixed the variance explosion of OLS in data-scarce, long-horizon settings.
- **Failure Mode Created:** We introduced bias. In highly stable, low-dimensional settings, Ridge/Lasso mathematically underperformed OLS.

Looking Ahead: Lecture 2

Next Session: Kernel Methods and SVMs

Lecture 2 will focus on Kernel Methods and Support Vector Machines (SVM) for Time Series.

Topics to prepare for:

- Maximum-margin learning and the Primal/Dual formulations.
- The "Kernel Trick" (RBF, Polynomial) to project linear boundaries into non-linear, high-dimensional spaces.
- Classification targets in macro/finance (e.g., recessions, stress events).

Reminder: Reading memos for the Cortes & Vapnik (1995) and Bergmeir et al. (2018) papers are due before the next Lecture!